DSA LAB-6
2.3.26
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1. Inserting a node at the beginning of a doubly-linklist:

Code-

```c
#include <stdio.h>
#include <stdlib.h>
struct Node {
    int data;
    struct Node *prev, *next;
};
void insertAtBeginning(struct Node** head, int val) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->data = val;
    newNode->next = *head;
    newNode->prev = NULL;
    if (*head != NULL) (*head)->prev = newNode;
    *head = newNode;
}
int main() {
    struct Node *n1 = (struct Node*)malloc(sizeof(struct Node));
    struct Node *n2 = (struct Node*)malloc(sizeof(struct Node));
    struct Node *n3 = (struct Node*)malloc(sizeof(struct Node));
    struct Node *n4 = (struct Node*)malloc(sizeof(struct Node));
    n1->data = 10; n1->prev = NULL; n1->next = n2;
    n2->data = 20; n2->prev = n1;   n2->next = n3;
    n3->data = 30; n3->prev = n2;   n3->next = n4;
    n4->data = 40; n4->prev = n3;   n4->next = NULL;
    struct Node* head = n1;
    insertAtBeginning(&head, 5);
    struct Node* temp = head;
    while(temp != NULL) {
        printf("%d <-> ", temp->data);
        temp = temp->next;
    }
    printf("NULL\n");
    return 0;
}
```

Output-

Output

```
5 <-> 10 <-> 20 <-> 30 <-> 40 <-> NULL
```

2.Inserting a node at the end of a doubly linked list:

Code-

```c
#include <stdio.h>
#include <stdlib.h>
struct Node {
    int data;
    struct Node *prev, *next;
};
void insertAtEnd(struct Node* head, int val) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->data = val;
    newNode->next = NULL;
    struct Node* temp = head;
    while(temp->next != NULL) temp = temp->next;
    temp->next = newNode;
    newNode->prev = temp;
}
int main() {
    struct Node *n1 = (struct Node*)malloc(sizeof(struct Node));
    struct Node *n2 = (struct Node*)malloc(sizeof(struct Node));
    struct Node *n3 = (struct Node*)malloc(sizeof(struct Node));
    struct Node *n4 = (struct Node*)malloc(sizeof(struct Node));
    n1->data = 10; n1->prev = NULL; n1->next = n2;
    n2->data = 20; n2->prev = n1;   n2->next = n3;
    n3->data = 30; n3->prev = n2;   n3->next = n4;
    n4->data = 40; n4->prev = n3;   n4->next = NULL;
    struct Node* head = n1;
    insertAtEnd(head, 50);
    struct Node* temp = head;
    while(temp != NULL) {
        printf("%d <-> ", temp->data);
        temp = temp->next;
    }
    printf("NULL\n");
    return 0;
}
```

Output-

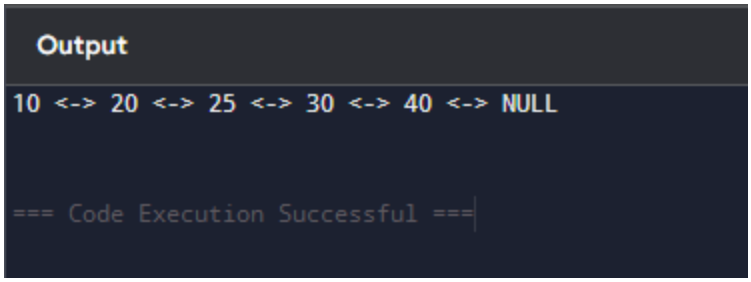
```
Output

10 <-> 20 <-> 30 <-> 40 <-> 50 <-> NULL


=== Code Execution Successful ===
```

3.Inserting a node in between a doubly link-list:(pos=3)

Code-

```c
#include <stdio.h>
#include <stdlib.h>
struct Node {
    int data;
    struct Node *prev, *next;
};
void insertAtPos3(struct Node* head, int val) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->data = val;
    struct Node* temp = head;
    temp = temp->next;
    newNode->next = temp->next;
    newNode->prev = temp;
    if(temp->next != NULL) temp->next->prev = newNode;
    temp->next = newNode;
}
int main() {
    struct Node *n1 = (struct Node*)malloc(sizeof(struct Node));
    struct Node *n2 = (struct Node*)malloc(sizeof(struct Node));
    struct Node *n3 = (struct Node*)malloc(sizeof(struct Node));
    struct Node *n4 = (struct Node*)malloc(sizeof(struct Node));
    n1->data = 10; n1->prev = NULL; n1->next = n2;
    n2->data = 20; n2->prev = n1;   n2->next = n3;
    n3->data = 30; n3->prev = n2;   n3->next = n4;
    n4->data = 40; n4->prev = n3;   n4->next = NULL;
    struct Node* head = n1;
    insertAtPos3(head, 25);
    struct Node* temp = head;
    while(temp != NULL) {
        printf("%d <-> ", temp->data);
        temp = temp->next;
    }
    printf("NULL\n");
    return 0;
}
```

Output-


Output
10 <-> 20 <-> 25 <-> 30 <-> 40 <-> NULL
=== Code Execution Successful ===